BRAC UNIVERSITY
Inspiring Excellence

CSE110: Programming Language I (Lab)
Spring 2026
Lab Exam - 02
Duration: 15 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

You are provided with a number N from the user. Your task is to print a pattern of numbers in descending order, where each row starts from n and decreases by 1 until the row number. The total number of rows will be equal to the value of N.

Sample Input	Sample Output
4	4 43 432 4321
6	6 65 654 6543 65432 654321

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B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

You are given a number N from the user. Your task is to print a pattern where the number of characters in each row increases by 1, and the symbols alternate between two characters. The pattern should continue for N rows.

Sample Input	Sample Output
4	& $$ &&& $$$$
6	& $$ &&& $$$$ &&&&& $$$$$$

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Lab Exam - 03
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

You are building a text filtering tool for a data cleaning application. Users often input strings where words are separated by a special character. Some characters in each word appear only once (considered noise), while others appear multiple times (considered significant).

Your task is to develop a program that removes all unique (single-occurrence) characters from each word while keeping all occurrences of characters that appear more than once.

The program should:

- Read a string containing multiple words separated by a user-specified character
- Read the separator character
- Split the string manually (no built-in split)
- For each word, remove characters that appear only once
- Print the filtered words

Input	Output
heellllloo-worrld -	The words are: eellllloo rr

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B

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

You are developing a text analyzer where users input strings with repeated characters and special separators. In this case, characters that appear only once are considered meaningful, while repeated characters are treated as noise.

Your task is to develop a program that keeps only the characters that appear exactly once in each word and removes all repeated characters.

The program should:

- Read a string containing multiple words separated by a user-specified character
- Read the separator character
- Split the string manually (no built-in split)
- For each word, keep only characters that appear once
- Print the cleaned words

Input	Output
heelllloo#worrld # 	The words are: h wold

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Spring 2025
Lab Exam - 04
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

You are given an integer array and you need to take a number as input from the user. Your task is to update the array by replacing the duplicate values of the array with zero. The updated array should be sorted in ascending order. Finally, you have to print the position of the number (you took as input from the user) from the updated array. Print **“Not found”** if the number is not in the array.

Can’t use built-in sort for the problem. Can’t create a new array for this problem.

Input	Output
int arr [] = {6, 2, 6, 1, 2, 9, 1} N=2	Updated array: 6, 2, 0, 1, 0, 9, 0 Sorted array: 0,0,0,1,2,6,9 Output: Position of 2 = 5
int arr [] = {6, 2, 6, 1, 2, 9, 1} N=5	Updated array: 6, 2, 0, 1, 0, 9, 0 Sorted array: 0,0,0,1,2,6,9 Output: Position of 5 = Not found

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B

Name:	ID:	Section:

Question 1 [CO3] [10 Points]

You are given an integer array and you need to take a number as input from the user. Your task is to create a new array that will contain only the unique (distinct) elements of the given array.The new array should be sorted in descending order. Finally, you have to print the position of the number (you took as input from the user) in the new array.Print **"Not found"** if the number is not in the array.

Can't use built-in sort for the problem.

Input	Output
int arr [] = {6, 2, 6, 1, 2, 9, 1} N=6	New array: 6, 2, 1, 9 Sorted array: 9,6,2,1 Output: Position of 6 = 2
int arr [] = {6, 2, 6, 1, 2, 9, 1} N=7	New array: 6, 2, 1, 9 Sorted array: 9,6,2,1 Output: Position of 7 = Not found